

Comparative Study of Machine Learning and Deep Learning Approaches for Mental Health Detection from Social Media Text

Final Year Project Proposal

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BS in CS



Department of CS

Fast School of Computing

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Project Registration

Project ID (for office use)	F25-112					
Type of project	☒ Traditional ☐ Industrial ☐ Continuing					
Nature of project	☐ Development ☐ Research & Development ☒ Research					
Sustainable Development Goals(SDGs)	☒ Good Health and Well-Being ☐ Quality Education ☐ Industry, Innovation, and Infrastructure ☐ Gender Equality ☐ Decent Work and Economic Growth ☐ Climate Action					
Area of specialization	☒ Artificial Intelligence (AI) ☒ Data Science and Analytics ☐ Internet of Things (IoT) ☐ Blockchain ☐ Mobile App Development ☐ Web Development ☐ Cybersecurity ☐ Game Development ☒ Natural Language Processing (NLP) ☐ Other					
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Declaration: FYP group members have cleared all prerequisite courses For FYP-I as per their degree requirements.						

Project Abstract

Mental health problems such as depression and anxiety are recognized as major global health concerns, affecting millions of people each year. Early detection and awareness can play a critical role in prevention and treatment. With the increasing use of social media platforms like Twitter and Reddit, individuals often share their feelings, struggles, and emotions online. These digital footprints create opportunities for computational approaches to identify potential indicators of mental health conditions.

This project aims to conduct a comparative study between classical machine learning (ML) models and deep learning (DL) models for detecting signs of mental health issues from social media text. ML methods such as Logistic Regression, Support Vector Machines, Random Forest, and Naïve Bayes will be tested against deep learning methods including Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), Convolutional Neural Networks (CNN), and transformer-based models such as BERT. The goal is to investigate which category of methods provides more accurate, interpretable, and practical results in detecting depression and anxiety.

The study will involve collecting publicly available datasets, preprocessing text data, implementing and training selected models, and evaluating performance using metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC. In addition to performance, factors such as training time, computational requirements, and interpretability will be analyzed. The findings will contribute to a structured benchmark that highlights the trade-offs between ML and DL approaches in this domain.

The expected outcome of this research is a comprehensive evaluation supported by experiments, graphs, and critical analysis. The project does not intend to serve as a clinical tool but rather as an academic contribution to computational mental health research, providing a reproducible study and insights for future work in this growing field.

1 Introduction

Mental health issues, including depression and anxiety, are becoming increasingly prevalent in today's society. Yet, they often remain undetected due to social stigma, lack of awareness, or delayed reporting. Social media platforms such as Twitter and Reddit provide a unique window into users' thoughts, emotions, and daily experiences, offering a rich source of textual data that can reveal early signs of mental health concerns. Traditional detection methods rely heavily on self-reporting and clinical assessments, which are often time-consuming, subjective, and reactive rather than proactive. With recent advances in Natural Language Processing (NLP) and machine learning, there is now an opportunity to analyze social media content automatically and detect mental health indicators at scale. Detecting these issues early can not only help individuals receive timely support but also provide large-scale insights into population mental health trends, aiding policymakers and healthcare providers in designing preventive strategies.

This project aims to develop a system that can identify signs of depression and anxiety from user-generated posts. The system will process large volumes of social media text, extract meaningful linguistic and semantic features using NLP techniques, and classify posts using machine learning models. In addition to classification, the project will explore the types of textual cues most indicative of mental health issues and evaluate how different model architectures impact detection performance. The system will also investigate how temporal patterns and user engagement behaviors correlate with mental health indicators, providing a more holistic understanding of online mental health expression. This approach has the potential to assist in early identification, support mental health professionals, and provide data-driven insights for research and intervention.

1.1 The main sub-tasks of this project include:

- Data Collection: Gathering social media posts while ensuring ethical use and protecting user privacy.
- Data Preprocessing: Cleaning and normalizing text, including tokenization, lemmatization, and removal of irrelevant content.
- Feature Extraction: Representing text using methods such as TF-IDF, word embeddings, and sentiment analysis.
- Model Development: Training and evaluating machine learning models, including Logistic Regression, Random Forest, and LSTM-based architectures.
- Evaluation: Assessing performance using metrics such as accuracy, precision, recall, and F1-score, and comparing results to baseline methods.

Success will be determined by the accuracy and reliability of the classification models, the robustness of the system across diverse datasets, and the insights gained from analyzing model predictions. Evidence for achieving these goals will be provided through implementation results, performance metrics, and comparisons with existing approaches. By combining NLP and machine learning, this project aims to provide an automated, scalable, and ethical framework for detecting mental health issues, ultimately contributing to early intervention, awareness, and research in the field of computational mental health. Furthermore, the findings of this project could serve as a foundation for future work in automated mental health support systems and policy-driven mental health initiatives.

2 Success Criterion

The success of this project will be determined by its ability to implement and evaluate a representative set of classical machine learning, deep learning, and transformer-based models for mental health detection on social media text. Success will include ensuring that all models are trained on the same preprocessed dataset, evaluated under a consistent framework, and compared using standard performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

The project will further be considered successful if the results are presented in a transparent manner through tables and visualizations, followed by a comparative analysis that highlights the strengths, weaknesses, and trade-offs of each approach. Beyond raw performance, success will also be reflected in the ability to provide meaningful insights into interpretability, computational efficiency, and practical applicability of different model families.

In summary, the project will be regarded as successful if it delivers a reproducible and balanced comparison across multiple modeling approaches, supported by clear evaluation and analysis that align with the objectives stated in the proposal.

3 Related work

Recent research at the intersection of natural language processing and mental health detection on social media has advanced rapidly, producing both large datasets and novel model architectures that motivate this project. Zhou and Mohd introduced a hybrid framework that combined preprocessing steps such as emoji normalization and slang replacement with transformer embeddings and a BiLSTM classifier, demonstrating that deep learning pipelines outperform traditional ML baselines when dealing with highly informal user-generated text [1]. Liu et al. conducted a systematic comparison of classical ML algorithms (Logistic Regression, SVM, Random Forest) against transformer-based models, finding that while transformers such as RoBERTa achieved the highest accuracy, classical approaches remained competitive where computational resources or interpretability were priorities [2].

Other researchers have emphasized ensemble and hybrid strategies. Tavchioski et al. explored ensembles of transformer models on Reddit and Twitter, showing that combining variants of BERT improved robustness and generalization compared to single-model approaches [3]. Similarly, Senanayake and Liang investigated anxiety detection in Reddit posts using emotion-based features and classical classifiers, reporting that Random Forest outperformed other methods with 89% accuracy, highlighting the enduring strength of feature-engineered ML approaches [4].

Recent work has also addressed multimodality and large-scale datasets. Mansoor and Ansari developed an AI-powered framework analyzing nearly one million multilingual social media posts, achieving 89.3% accuracy for early detection of mental health crises and demonstrating the importance of scaling across languages and platforms [5]. Sharma et al. proposed NeuroVibeNet, integrating behavioral and voice data with ensemble ML classifiers such as Random Forest, LightGBM, and SVM, reporting near-perfect performance (~99%), underscoring the benefits of multimodal analysis [6].

Comparative and review studies further highlight emerging trends. Ding et al. systematically examined trade-offs between ML and DL approaches, noting that while DL excels in accuracy, ML remains advantageous in interpretability and efficiency [7]. Other surveys stress the role of preprocessing, dataset imbalance handling, and emotion-based features as consistent determinants of model success [8].

Taken together, these works show (1) the dominance of transformers and hybrid deep learning methods in recent literature, (2) the enduring role of classical ML where transparency or limited resources are important, and (3) promising directions in multimodality and large-scale multilingual analysis. However, most existing research either focuses solely on one paradigm (ML or DL) or compares them across inconsistent preprocessing pipelines. By systematically implementing and evaluating both classical ML and deep learning models under a unified framework, this project directly addresses that gap.

4 Project Rationale

The growing reliance on social media for self-expression has made it an important source for detecting early signs of mental health issues, yet distinguishing genuine signals of distress from the vast and noisy text data remains a major challenge. This project is motivated by the need to explore how different computational approaches—ranging from classical machine learning to modern deep learning and transformer-based models—perform in identifying such patterns. By undertaking this research, we aim not only to evaluate the relative strengths and limitations of these models but also to contribute insights into how accuracy, interpretability, and computational efficiency can be balanced in sensitive applications like mental health detection. The purpose of this work is to develop a fair and reproducible comparison across model families, which will help guide future research and practical applications in AI for healthcare, while giving us hands-on experience in applying advanced machine learning techniques to socially relevant problems.

4.1 Aims and Objectives

The aim of this project is to investigate and compare the effectiveness of classical machine learning, deep learning, and transformer-based models for detecting mental health conditions from social media text. By developing a unified experimental framework, the project seeks to provide a balanced evaluation of different modeling approaches in terms of performance, interpretability, and computational efficiency.

To achieve this aim, the project will focus on the following objectives:

- To collect and preprocess a suitable dataset of social media text for mental health detection.
- To implement representative classical machine learning models for baseline performance evaluation.
- To design and train deep learning models and fine-tune transformer-based models for benchmarking against state-of-the-art methods.
- To evaluate all models using standard metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

- To conduct a comparative analysis highlighting trade-offs in accuracy, interpretability, and computational requirements.

4.2 Scope of the Project

The scope of this project is limited to the implementation and evaluation of selected classical machine learning, deep learning, and transformer-based models for the detection of mental health conditions from social media text. The work will include dataset collection and preprocessing, feature extraction, model implementation, training, evaluation, and comparative analysis. The primary deliverables will consist of trained models, performance evaluation results presented through tables and graphs, and a comprehensive report documenting the methodology, experiments, and findings.

The scope does not extend to the development of a deployable application, real-time monitoring system, or large-scale clinical validation, as the focus is strictly on comparative research. Deadlines will follow the academic calendar, with major milestones including dataset preparation, model implementation, evaluation, and report submission. Costs are expected to be minimal, limited to computing resources and open-source libraries.

5 Proposed Methodology and Architecture

The methodology adopted for this project follows a structured workflow designed to ensure fairness and consistency in the comparison of classical machine learning, deep learning, and transformer-based models. The process begins with dataset preparation, where social media text related to mental health will be collected and subjected to preprocessing steps such as cleaning, normalization, tokenization, and stopword removal. These steps are necessary to reduce noise and standardize the text for subsequent analysis.

Following preprocessing, feature engineering techniques will be applied to convert text into meaningful numerical representations. Approaches such as TF-IDF, Word2Vec, and GloVe will be used to ensure compatibility with different model families. Based on these features, models from three categories will be developed and evaluated: classical machine learning models (Logistic Regression, Naïve Bayes, Support Vector Machine, Random Forest), deep learning models (CNN, LSTM, BiLSTM), and transformer-based models (BERT).

All models will be trained and tested under a unified framework, using the same preprocessed dataset and evaluation criteria. Performance will be measured using standard metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Finally, a comparative analysis will be conducted to highlight the strengths, weaknesses, and trade-offs among the models, particularly in terms of accuracy, interpretability, and computational efficiency. This approach ensures that the project not only delivers reliable performance comparisons but also generates meaningful insights for future research.

6 Individual Tasks

Team Member	Activity	Tentative Date
Tahir Haider	Conduct literature review on classical ML, DL, and transformer-based approaches for mental health detection.	Aug – Sep 2025
	Collect and preprocess dataset (text normalization, tokenization, stopword removal, handling missing values).	Aug – Sep 2025
	Implement transformer models (BERT, optional RoBERTa) using Hugging Face library.	Jan – Feb 2026
	Comparative evaluation of transformers against classical ML and DL models.	Feb – Mar 2026
	Contribute to final report writing and preparation of presentation slides.	Apr – May 2026
M. Huzaifa Saleem	Extract features (TF-IDF, Bag-of-Words) for classical ML models.	Oct 2025
	Implement classical ML models (Logistic Regression, Naïve Bayes, SVM, Random Forest).	Oct – Nov 2025
	Evaluate classical ML models and establish baseline performance.	Nov 2025
	Lead preparation of results tables, graphs, and baseline comparison analysis.	Feb – Mar 2026
	Contribute to documentation of methodology and result interpretation.	Mar – Apr 2026
S.M. Sufyan	Build word embeddings (Word2Vec, GloVe) for deep learning models.	Nov 2025
	Implement CNN, LSTM, and BiLSTM models.	Nov – Jan 2026
	Perform hyperparameter tuning for deep learning models.	Dec – Jan 2026
	Draft discussion and analysis section (trade-offs: accuracy vs interpretability vs computational cost).	Mar – Apr 2026
	Contribute to final proofreading and defense presentation preparation.	Apr – May 2026

Table 1. Individual Tasks

7 Gantt Chart

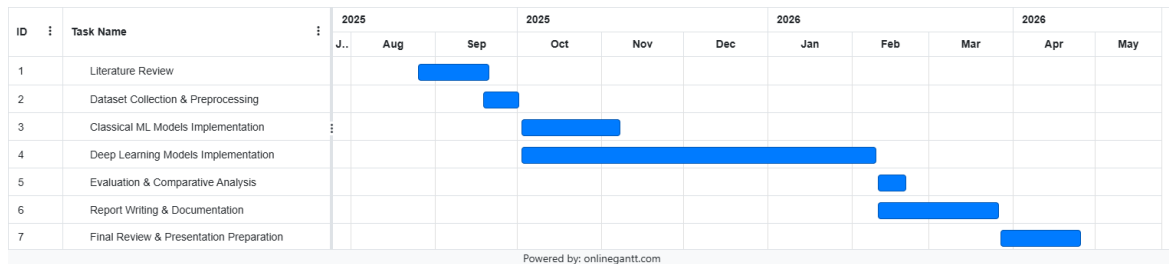


Figure 1. Gantt Chart

8 Tools and Technologies

The project will make use of a range of tools and technologies to support dataset preprocessing, model implementation, training, and evaluation.

8.1 Programming Language

- **Python** will be the primary programming language due to its extensive ecosystem of machine learning and deep learning libraries.

8.2 Libraries and Frameworks:

- **Scikit-learn** for implementing classical ML models such as Logistic Regression, Naïve Bayes, SVM, and Random Forest.
- **TensorFlow** and **Keras** for building and training deep learning models such as CNN, LSTM, and BiLSTM.
- **PyTorch** and **Hugging Face Transformers** for fine-tuning transformer-based models like BERT.
- **NLTK** and **spaCy** for text preprocessing tasks, including tokenization, stopword removal, and stemming/lemmatization.

8.3 Data Handling and Analysis:

- **Pandas** and **NumPy** for dataset manipulation and numerical computations.
- **Matplotlib** and **Seaborn** for visualizing results in the form of graphs and charts.

8.4 Development Environment:

- **Jupyter Notebook** and **VS Code** for coding, experimentation, and documentation.

8.5 Version Control:

- **Git** and **GitHub** for collaborative development and version management.

8.6 Project Management:

- **Jira** (or equivalent) for task management and progress tracking.

9 References

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